Quiz 5 - Results

Attempt 1 of 1

Written Oct 20, 2026 10:57 AM - Oct 20, 2026 11:02 AM

Attempt Score	**8 / 10 - 80 %**
Overall Grade (Highest Attempt)	**8 / 10 - 80 %**

Question 1 — 2 / 2 points

In the knapsack problem without repetition, how many times can an item be chosen?

- ○ Any number of times
- ○ Exactly once
- ✔ ◉ At most once
- ○ Only if it is the most valuable item

Question 2 — 0 / 2 points

Matrix A is (10 \times 5), B is (5 \times 2), and C is (2 \times 10). Using standard matrix multiplication, which statement is correct?

- ○ (A(BC)) requires 300 scalar multiplications and is cheaper.
- ○ ((AB)C) requires 600 scalar multiplications and is cheaper.
- ➡ ○ ((AB)C) requires 300 scalar multiplications and is cheaper.
- ✖ ◉ Both ((AB)C) and (A(BC)) require 300 scalar multiplications

Question 3

2 / 2 points

A knapsack has a capacity of 6. The available item types are:

Item	Weight	Value
A	3	5
B	4	7

How does the maximum value change when unlimited copies are allowed instead of at most one copy of each item?

- ○ It increases from 5 to 10.
- ○ It stays at 7.
- ✓ ◉ It increases from 7 to 10.
- ○ It increases from 7 to 12.

Question 4

2 / 2 points

What is the goal of the knapsack problem?

- ✓ ◉ Choose the most valuable items without exceeding the weight limit
- ○ Make the bag as heavy as possible
- ○ Choose the lightest item
- ○ Put every item in the bag

Question 5

2 / 2 points

In 0/1 knapsack, (K(w,j)) is the best value using only the first (j) items with capacity (w). If item (j) is selected, why do we use (K(w-w_j,j-1)+v_j)?

- ○ To ensure that item (j-1) is also selected.
- ○ To ensure that the bag is filled exactly.
- ○ To remove the most valuable remaining item.
- ✓ ◉ To prevent item (j) from being selected again.

Done